Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Physicist with a Ph.D. in Physics and over ten years of research experience in computational astrophysics, numerical methods, and high-performance computing. One term of classroom instruction at the graduate level and multi-year experience mentoring graduate researchers through the transition from confusion to independent work. Teaching approach built on intuition and critical thinking over memorization, and on collaborative problem-solving.

Teaching and Mentoring

Thermodynamics (graduate level) **Jun 2018**
Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Substitute Instructor

Zachary Davis, graduate student **2018 – 2022**
Department of Physics and Astronomy, Purdue University Mentoring

Amanda O. Harrison, graduate student **2018 – 2020**
Department of Physics and Astronomy, Purdue University Mentoring

Hao Zhang, graduate student **2018 – 2019**
Department of Physics and Astronomy, Purdue University Mentoring

Education

Universitat de València **Jul 2017**
Ph.D. in Physics, Excellent Cum Laude

Universidad Michoacana de San Nicolás de Hidalgo **Sep 2011**
M.Sc. in Physics

Universidad Autónoma del Estado de México **Dec 2008**
B.Sc. in Physics

Experience

Independent Contractor **Jan 2026 – Present**
Mercor
- Evaluate physics reasoning in AI model outputs for technical accuracy, identifying failure modes and providing structured feedback.

Postdoctoral Research Fellow **Oct 2018 – Nov 2020**
Purdue University
- Obtained and led a NASA-funded grant studying radiation from active galaxies.
- Mentored three graduate researchers; two are now postdoctoral researchers.
- Published 3 peer-reviewed papers.

Postdoctoral Research Associate **Feb 2021 – Apr 2022**
Rochester Institute of Technology
- Led an NSF-sponsored project upgrading a high-performance simulation code for modeling black hole binaries.
- Collaborated on a multi-institutional study of binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow **Jan – Sep 2018**
Universidad Michoacana de San Nicolás de Hidalgo
- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Skills

Physics: Mechanics, electromagnetism, thermodynamics, relativity, statistical physics, astrophysics

Computing: Python, Fortran, C/C++, high-performance/parallel computing

Languages: English (fluent), Spanish (native), Nahuatl (basic)

Selected Publications (9 total, 3 first-author)

Full list available on request.